TIDYWELLS REFERENCE SHEET

Units & Conversions

The factors Canadian oil and gas data keeps tripping people on.

Oil

FROM	TO
1 m^3 oil	6.2898 bbl
1 bbl	0.158987 m^3
1 m^3/d	6.2898 bbl/d
1,000 m^3 oil	6,289.8 bbl

Gas

FROM	TO
1 m^3 gas	35.3147 ft^3
1 10^3m^3 (e3m3)	35.3147 Mcf
1 Mcf	1,000 ft^3
1 MMcf	1,000 Mcf
1 Bcf	1,000 MMcf

Energy & equivalents

FROM	TO
1 GJ	0.947817 MMBtu
1 MMBtu	1.05506 GJ
6 Mcf gas	1 boe (6:1 rule of thumb)

The 6:1 boe ratio is an energy convention, not a price or a physical law. Gas heating value varies by stream, so GJ per m^3 is not constant.

Pressure

FROM	TO
1 kPa	0.145038 psi
1 psi	6.89476 kPa
1 MPa	1,000 kPa

From monthly volume to a daily rate

```
rate = monthly volume ÷ days in that month
```

Regulators publish **monthly volumes**. Compare months by rate, not volume, or February will always look like a collapse. Alberta and Saskatchewan (Petrinex) report oil in m^3 and gas in 10^3m^3; BC Energy Regulator files use their own units, so check the header before you add anything up.

```
AB oil: m³ × 6.2898 ÷ 1000 = thousand bbl for the month
AB gas: 10³m³ × 35.3147 ÷ 1000 = MMcf for the month
```

Prefix traps

- **M** means thousand in field use (Mcf), but **MM** means million. In SI, M is mega (million). Always check which system you are reading.
- **k** is thousand: 1 kbbl/d = 1,000 bbl/d. **e3m3** and **10^3m^3** are the same unit.
- Cumulative volumes are not rates. Never average them.

Standard published conversion factors. Check units in the source file before relying on any result. Not engineering or investment advice.